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PLANETARY SCIENCE

Reorientation and despinning of 4 Vesta formed the Divalia Fossae

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Vesta, the only differentiated rocky protoplanet explored by a spacecraft, offers insight into early planetary formation. The Divalia Fossae, surface troughs comparable in size to the Grand Canyon, encircle two-thirds of the equator. Two giant impacts reshaped the southern hemisphere, where an older basin is partially superposed by the younger Rheasilvia basin. The origin of the Divalia Fossae is widely accepted as directly linked to the Rheasilvia impact, either by tectonics caused immediately by the impact, up-spinning, or secondary cratering. We present several geologic constraints that support a tectonic origin of the troughs due to the adjustment of Vesta's spin axis to a geoid changed by both large impacts. The best fit to Vesta's gravitational field corresponds to a spin axis reorientation of 3° that, when coupled with despinning, induces a stress state that predicts Divalia Fossae's established location, fracture type, and orientation. These insights underline the importance of tectonic processes in the early evolution of protoplanets.

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INTRODUCTION

Vesta is spinning rapidly with a well-determined rotational period of 5.342 hours, and it has a marked equatorial bulge and polar flattening (Fig. 1A). Its shape is also controlled by hemisphere-scale impact sites with the Veneneia basin being partially superposed by the Rheasilvia basin, together forming a depression that occupies the majority of the southern hemisphere (Fig. 1B). The impact ejecta from these basin-forming events is thought to be the source for the howardite-eucrite-diogenite meteorites (*1*). Vesta hosts two distinct sets of large-scale troughs bounded by steep scarps (Fig. 1A). The Divalia Fossae are situated on Vesta's equatorial bulge, just south of the equator in roughly east-west orientations. The Saturnalia Fossae, oriented northwest-southeast, are only exposed in the northern hemisphere with their southern extent truncated by the Divalia Fossae (*2*).

There are multiple hypotheses that have been proposed for the formation of Vesta's troughs (*2*–*5*). The hypothesis that found the most attention suggests that, although they are unusual in size and morphology for tectonic structures on small bodies, these troughs are graben directly formed by the Rheasilvia impact via normal faulting (*2*, *6*). The same process is invoked for the formation of the Saturnalia Fossae by the Veneneia impact. While multiple numerical models (*7*, *8*) investigated the effects of giant impacts on the formation of troughs on Vesta, this scenario is largely motivated by the observation that the poles of the planes delineated by the Divalia Fossae are clustered at or near the center of the Rheasilvia basin (*2*), plausibly explained the formation of the Divalia Fossae directly by, and simultaneously with the Rheasilvia impact.

In our previous work (*9*), we reassessed the geographic relationship of the Divalia Fossae and Rheasilvia basin center. We defined the geometric mean center of the Rheasilvia basin (Fig. 2), which accounts for the irregularity of the basin rim, and compared it with the poles of planes through the troughs of Divalia Fossae. The geometry mean center of the Rheasilvia basin was found to lie outside of the 95% confidence interval of these poles (see Fig. 2) and thus clearly showed that the Divalia Fossae are not concentric around the Rheasilvia basin center (*9*). While the center of an impact basin may not precisely correspond to the impact point, especially when the impact is oblique, this reassessment challenges the hypothesis that the formation of the Divalia Fossae was immediately triggered by the Rheasilvia impact event.

Moreover, crater counting (*10*) and the fact that the Divalia Fossae crosscut the Rheasilvia basin (*9*) together provide unambiguous evidence that the troughs must have formed well after the basin emplacement and modification. The impact basin modification is estimated to have taken place approximately 2 to 3 hours after the impact (*7*, *11*, *12*). For typical earthquake rupture propagation rates, the Divalia Fossae, if indeed directly caused by the Rheasilvia impact event, would have formed before or within the same timeframe of the modification stage of the Rheasilvia basin, contradicting the cross-cutting relationship.

Furthermore, a large impact near the south pole is unlikely to induce deformation only along the equator without producing any noticeable effects at the antipode (*13*, *14*). While the region surrounding the antipode of the Rheasilvia impact is topographically elevated, the actual site of the antipode is superposed by a 90-km-diameter impact basin that together with the low-resolution of Dawn imagery at Vesta's north pole renders any evidence of antipodal deformation ambiguous (*14*).

Recent studies proposed that the Divalia Fossae could be secondary crater chains produced by the Rheasilvia impact (*4*, *5*). Modeling secondary cratering from the Rheasilvia impact predicts that latitudinal trough-like structures can form one of many possible sets of secondary cater chains (*4*). However, a cratering origin does not account for the systematic trough width distribution, where troughs are widest in the center with widths tapering towards trough terminations, a characteristic typical for tectonic structures (*15*). Furthermore, the Divalia Fossae are associated with subparallel pit crater chains that have morphologies inconsistent with those formed by secondary impact craters but consistent with a tectonic origin (*6*) that are comparable to the pit chains on multiple other planetary bodies (*16*–*20*).

Because our geologic observations refute trough formation directly by a large impact and do not align with an origin by secondary

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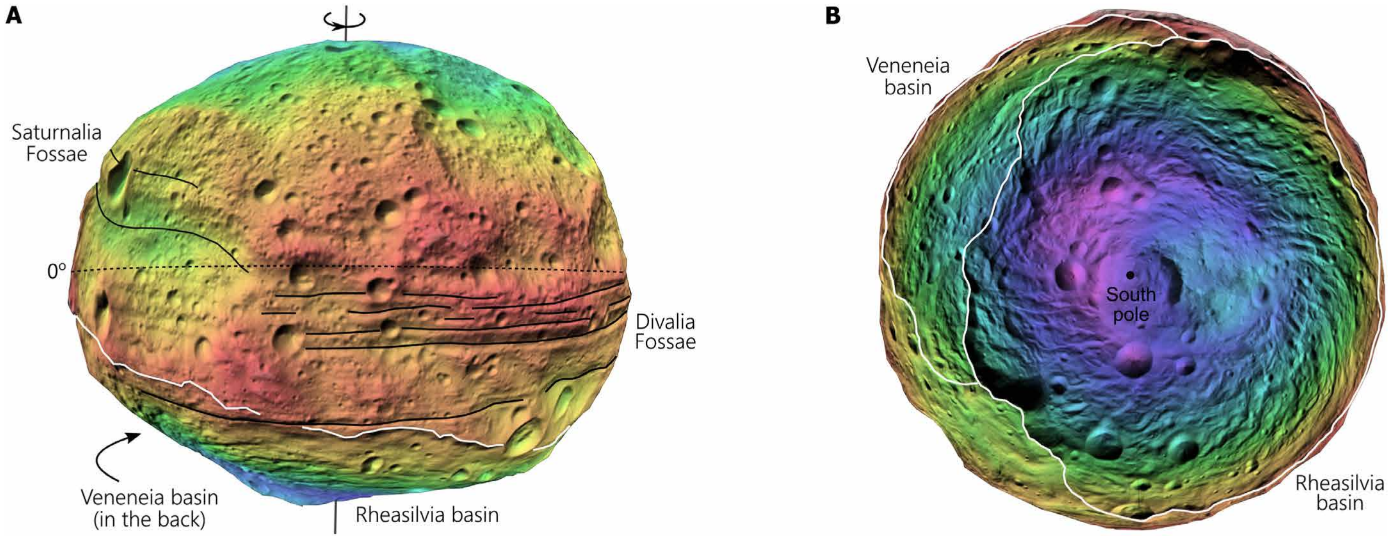

Fig. 1. Shape model of Vesta showing the two major sets of troughs and the south polar depression formed by two impact basins. (**A**) Equatorial view of Vesta showing major landforms with respect to spin axis and equator. (**B**) Vesta's southern hemisphere centered on the south pole. Elevations are color coded with warmer tones representing higher elevations and cooler tones representing lower elevations. The large-scale troughs, Divalia and Saturnalia Fossae, are labeled with black lines. The outlines of the Rheasilvia and Veneneia basins are marked with white lines. Three dimensional (3D) shape model is from the Vesta Trek Portal (*45*).

cratering, we investigate a scenario satisfying all geologic constraints that are related to Vesta's spinning history, especially given that the Divalia Fossae are found to be concentric around the spin axis (*9*).

RESULTS

The Veneneia and Rheasilvia basins together form an enormous depression covering the majority of the southern hemisphere (Fig. 2). We calculated the location of the geometric mean center of this depression, defined by the rims of the two basins, and it coincides not only with the location of the spin axis at the south pole but also falls near the topographically lowest point of the depression (Fig. 2). We compare the locations of the poles of planes of the Divalia Fossae (*9*) to the geometric mean center of the depression. The poles are colocated within 95% or higher confidence and so, too, coincide with the south pole and center of the depression formed by the Veneneia and Rheasilvia basins (Fig. 2). This establishes a well-defined geographic relationship between the depression formed by the two impact structures and the spin axis. We thus hypothesize and further investigate the origin of the Divalia Fossae due to a change in Vesta's spin caused by geoid adjustments resulting from the depression formed by the two large south polar impact structures.

Large impacts have the potential to disturb the rotational state of a planetary body by altering its inertia tensor, ultimately leading the planet to transition to a different minimum-energy state characterized by principal axis rotation (*21*). These changes encompass the migration of the rotation pole relative to the planet's surface, commonly known as true polar wander (TPW), and variations in the rotation rate (*21*). Before exploration by spacecraft, Vesta's gravity model, stress distribution, and tectonic patterns were predicted for combinations of reorientation and change in spin rate (*22*). With the availability of spacecraft observations of images (*23*), topography and shape model (*24*), and gravity model (*25*), as well as the collection of geologic observations providing tight boundary conditions and controls for any hypothesis, it is possible to specifically test if a change in spinning behavior formed the Divalia Fossae. We determined optimal solutions for TPW, despinning, and up-spinning by using the observed gravity field as a constraint (*21*). Initially, we derived best-fit solutions for the gravity anomaly due to Veneneia and Rheasilvia using gravity data limited to spherical harmonic degree-3 and higher, as rotational contributions dominate at degree-2 (see Materials and Methods). Once obtaining optimal models for Veneneia and Rheasilvia, we incorporated the degree-2 gravity field to find best-fit solutions for TPW, despinning, and up-spinning, which include rotational contributions (see Materials and Methods). The resulting best-fit solution indicates an initial rotation pole at 87°N 41°E and 16% despinning to the present rotation rate. The best-fit value of despinning exceeds the ~6% value calculated by (*26*) but is consistent with the larger value estimated by (*27*). We also note that while large impacts are known to have the potential to change the spin rate of planetary bodies, the slowing of the spin rate in this study is considered high (*3*), and lower values of despinning would still result in a similar stress distribution. With this solution in hand, we computed corresponding global stresses (*28*).

We then analyzed the stress field using rock-mechanics theory to identify locations where rock failure takes place and predict fracture type, as well as optimal and ranges of permissible fracture orientations (see method in the Supplementary Materials). These predictions were then compared to the location, orientation, and fracture type of the Divalia Fossae. Divalia Fossae are roughly trending east-west between longitudes ~110°E and 320°E, but they are found not to form a perfect great circle in the equatorial plane (*9*). While earlier studies assumed that the troughs are grabens (*6*), their geomorphology paired with rock mechanical calculations accounting for Vesta's low-gravity conditions, suggest that they are joints, accommodating mostly opening displacement (*15*).

We find that the best-fit solution obtained using the gravity field as a constraint produces a stress field that satisfies all geologic

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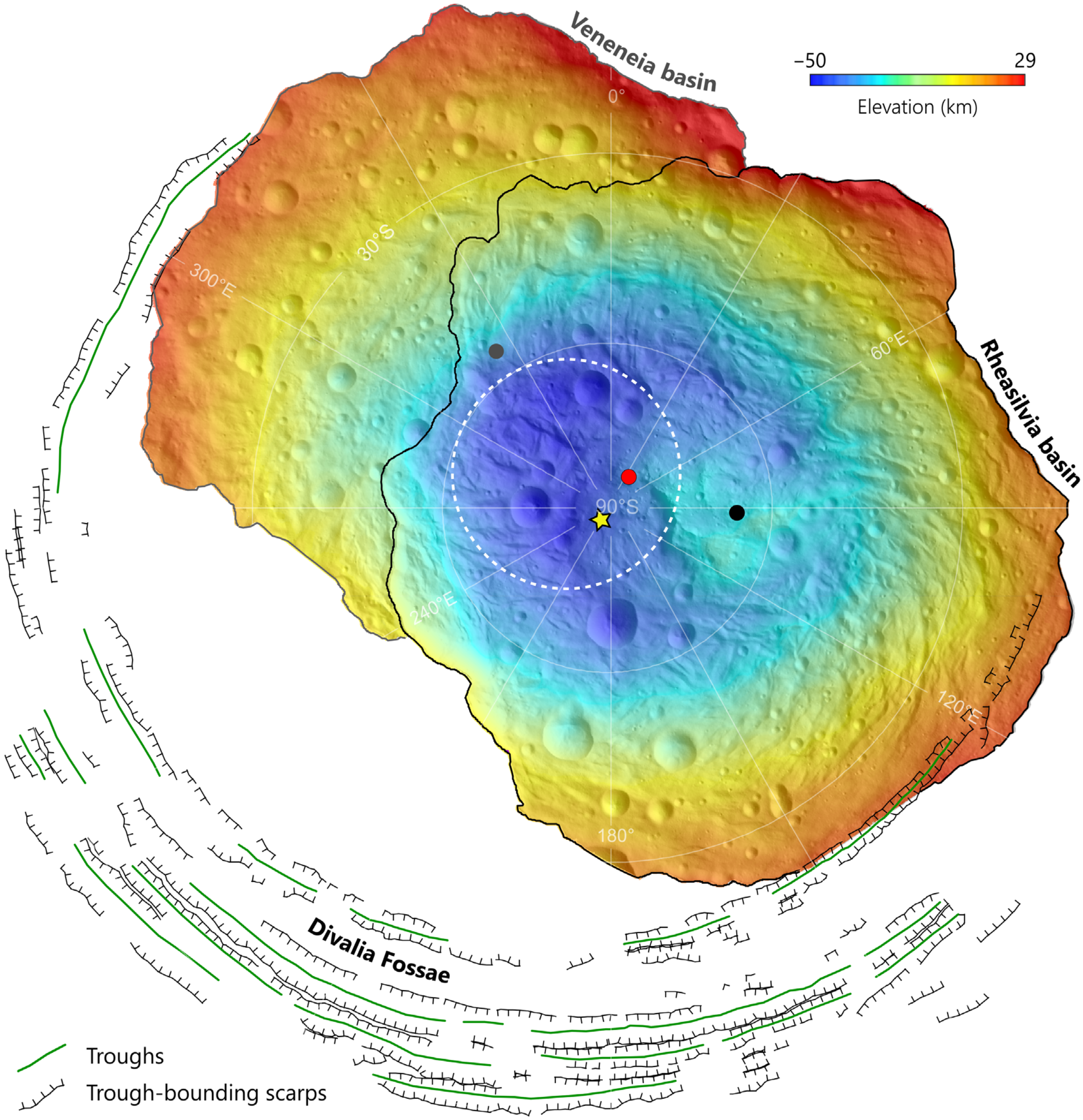

Fig. 2. The geographic relationship between the basins, troughs, and spin axis of Vesta in south polar stereographic projection. Divalia Fossae mapped with the troughs as solid green lines and the bounding scarps as black lines shown with respect to the topographic depression formed by the Veneneia and Rheasilvia basins that are color-coded by elevation. The basin-bounding scarps of the Rheasilvia and Veneneia basins are outlined in black and grey, respectively, their combined geometric mean center is located at 83.6°S and 34.4°E (red dot). The geometric mean center of the Rheasilvia basin alone is located at 69.3°S and 95.5°E (black dot), and that of the Veneneia basin is located at 56.3°S and 322.9°E (gray dot). The elevation map is with reference to the mean planetary radius of 262 km (*24*). The ellipse corresponding to the 95% confidence interval of the poles of Divalia Fossae is outlined with a white dotted line (*9*). The yellow star indicates the initial rotation pole at 87°S 221°E in the modeling presented in Fig. 3.

observations. The resulting global stress distribution (Fig. 3A) shows the type (Fig. 3A, color of lines), magnitude (Fig. 3A, length of lines), and orientation of the horizontal principal stress components. The induced stresses from this scenario systematically vary in orientation, showing radial and concentric orientations around the paleopoles at 87°N 41°E and 87°S 221°E. Stresses are entirely tensile in the polar regions. At the equator and toward the mid-latitudes, they are tensile in roughly north-south orientations and compressive in roughly east-west orientations. We note that our model presents only one but not the only solution for the stress magnitudes, as they are dependent on the lithospheric parameters used in the methods (see Materials and Methods). However, the mode of stress (i.e., compressive versus tensile) and the relative stress magnitudes will not be affected by different lithospheric parameters in the same TPW and despinning scenario and thus, will not change the tectonic prediction in our modeling.

For this stress state, the full set of rock-mechanical predictions of fracture types, orientations, and locations are visualized (Fig. 3B), and the analytical solutions are provided in the Supplementary

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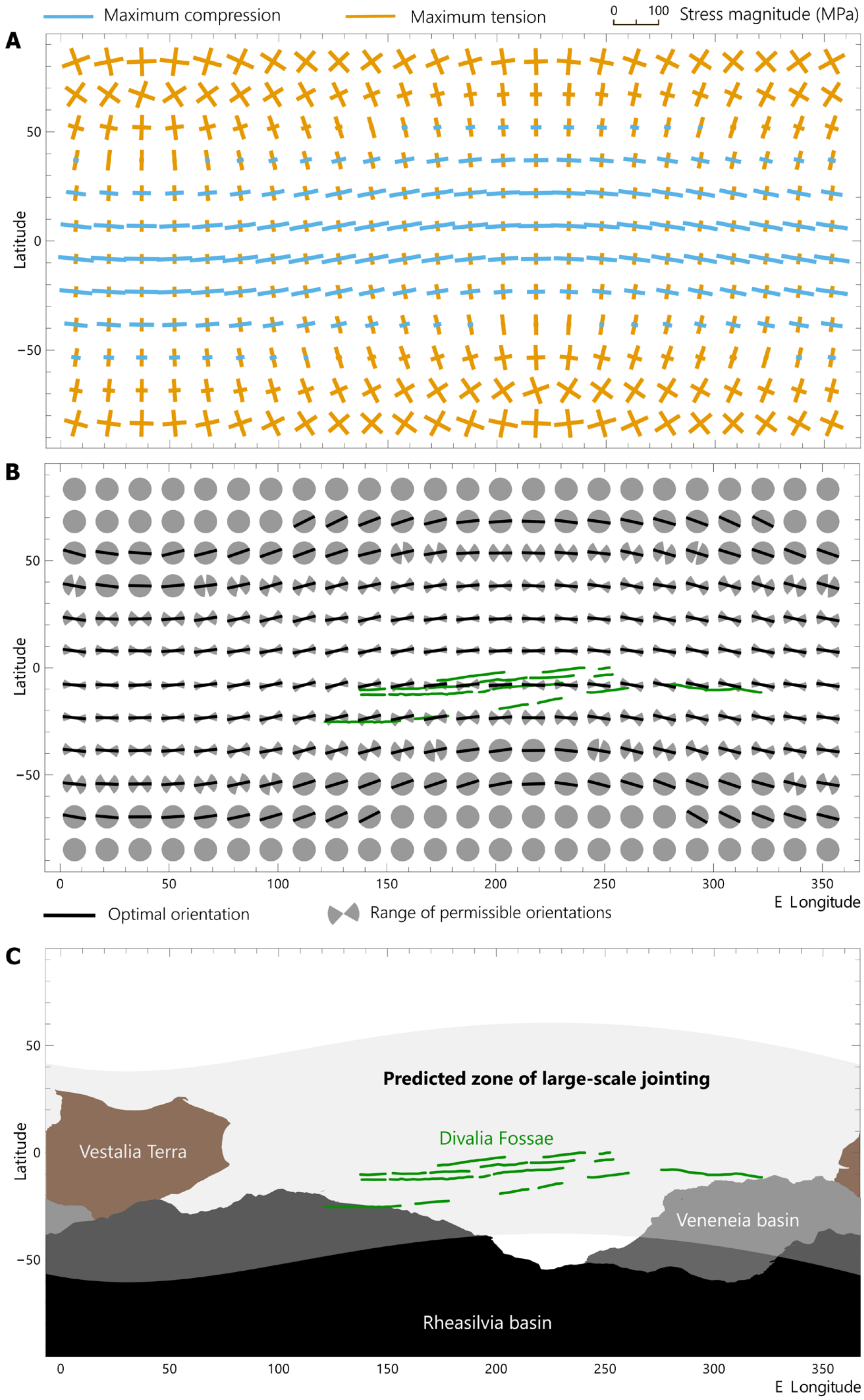

Fig. 3. Tectonic predictions of reorientation and despinning on Vesta. (**A**) Stress distribution showing the horizontal maximum and minimum principal stress components. The relative magnitude and orientations of the stresses are indicated by the length and color of the line symbols, with compressive stresses shown with blue lines and tensile stresses with orange lines. (**B**) Predicted global tectonic pattern from the global stress distribution in (A). Optimal orientations of fractures are shown as black lines, and permissible ranges in fracture orientation are marked in grey. The Divalia Fossae are outlined with green lines. Maps are shown in simple cylindrical projections, all calculations are plotted in geographic bins of 15° by 15°. The analytical solutions are provided in data S1. (**C**) Global map showing the Rheasilvia basin (black), Veneneia basin (gray), Vestalia Terra (brown), and Divalia Fossae (green lines) with the area where large-scale jointing is predicted shaded in gray on top.

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Materials. The map pattern of the Divalia Fossae is plotted for comparison. Because sufficiently large tensile stresses are found across the entire globe in this stress state, joints are the preferred fracture type, and rock failure via jointing is favorable globally. Any type of faulting, including normal faulting that produces graben, would only occur in a purely compressive stress state (*29*), which is not predicted anywhere on the surface of Vesta in the models (Fig. 3A). Thus, the predicted fracturing type matches that of our previous finding that the Divalia Fossae are joints (*15*).

At the equator, the average tensile stresses predicted in our model range from 12 to 26 MPa, exceeding the tensile strength of basaltic rock mass (*30*). Therefore, joints are predicted to be oriented generally east-west but also following the radial and concentric patterns around the paleopoles that are expressed most subtly at these latitudes. All of the Divalia Fossae are consistent with the range of predicted permissible orientations of joints but, in fact, the majority of orientations are at or near optimal orientations (Fig. 3B; also see fig. S1), closely matching observations and predictions. In addition, the predicted radial and concentric pattern around the paleopoles is in agreement with the Divalia Fossae not forming a perfect great circle in the equatorial plane (*9*).

Although failure is also predicted in the equatorial regions between longitudes 320°E and 110°E, the Divalia Fossae are absent there. This region corresponds to Vestalia Terra (Fig. 3C), an ancient plateau that is found to exhibit greater mechanical strength as compared to the rest of Vesta (*31*, *32*). This strength contrast is likely to account for the absence of the Divalia Fossae in this region. This observed absence is not only in support of models of a tectonic origin of the Divalia Fossae but further questions the secondary impact hypotheses (*4*, *5*), as a chain of secondary impacts would be expected to continue onto Vestalia Terra.

Failure via jointing is predicted to form in the higher latitudes and at the poles (Fig. 3B), but no large-scale joints are observed in these regions. This may be the case because the fracturing associated with the Divalia Fossae may have released stresses in the lithosphere and thus prevented fracturing in higher latitudes, a process that is generally unaccounted for in elastic models. Furthermore, the horizontal principal stresses here are purely tensile and nearly isotropic, a situation that promotes joints of all orientations with no strongly preferred alignments (Fig. 3B, gray circles). This leads to smaller and more distributed joints. Lineaments are abundant and widely distributed on the Rheasilvia basin floor in the south polar region, and they show no preferred orientations (*9*). However, their small sizes prevent clear identification of their fracture type in the available image datasets. Joints or any small lineaments similar to those in the south polar regions are not observed in the available Dawn images of the north polar regions. Furthermore, joints that are much smaller than those forming the troughs of the Divalia and Saturnalia Fossae are not expected to produce pronounced topographic landforms and thus may remain hidden under a regolith cover.

The Divalia Fossae developed into such large landforms due to the horizontal principal stress state being mixed, i.e., showing compressive and tensile components, and being highly anisotropic (Fig. 3A). Such a stress state promotes stronger alignments of joints along the optimal fracture orientation (Fig. 3B), which also increases opening displacements only along one prominent orientation at the equatorial region (Fig. 3C). This may have served to grow the Divalia Fossae into bigger landforms that are resolved in image data and where processes of mass wasting and regolith formation enhanced the width of the landforms (*15*).

DISCUSSION

Calculations of changes in the spinning behavior that are strongly supported by geological evidence allow us to reinterpret the geologic evolution of Vesta, involving TPW and despinning. The idea of reorientation of Vesta was previously investigated but discarded, because the lithosphere is predicted to be not compliant enough to perform efficient relaxation of impact basins and the rotational bulge (*33*). Our model takes into account the elastic support of the basins and a remnant rotational bulge linked to the inferred TPW and despinning, which is also sustained by the elastic lithosphere (*21*) (see Materials and Methods).

An origin of the Divalia Fossae from TPW and despinning requires reinterpretation of the tectonic history of Vesta. When the impact that formed the Rheasilvia basin hit Vesta near its south pole (Fig. 4A) it increased the size of the topographic depression formed by the preexisting Veneneia basin (Fig. 4B), creating a negative geoid anomaly. The adjustment in geoid caused the asteroid to reorient itself and a slowdown in rotation (Fig. 4C). These processes together served to induce a stress field across the globe (Fig. 3A) that was conducive to forming the Divalia Fossae via jointing along the equatorial budge (Fig. 4D).

It is plausible that a similar origin can be invoked for the Saturnalia Fossae and Veneneia impact at an earlier stage in Vesta's geologic evolution. There is limited geologic and geophysical information

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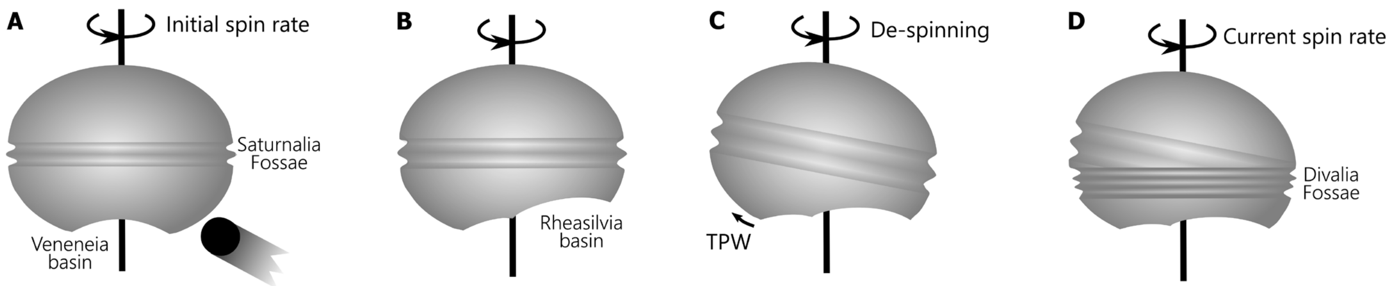

Fig. 4. Conceptual diagrams showing the tectonic evolution of Vesta. (**A**) The Veneneia basin is located near the south pole of Vesta with the Saturnalia Fossae also present at unknown orientations. A large impact occurred near the south pole. (**B**) The Rheasilvia basin is emplaced, partly superposing the Veneneia basin. (**C**) True polar wander (TPW) and despinning occur. (**D**) As a consequence of (C), the Divalia Fossae form near the equator and crosscut the Saturnalia Fossae, which are now only exposed in the northern hemisphere.

preserved, such that details on the degree of TPW or whether a change in spin rate occurred remain unknown, but reorientation of the asteroid from mass redistribution caused by the formation of the Veneneia basin may have triggered the fracturing associated with the Saturnalia Fossae (Fig. 4A). These troughs then degraded and widened (Fig. 4B), reoriented (Fig. 4C), until they were crosscut by the Divalia Fossae (Fig. 4D), forming the observed configuration of tectonic structures.

This tectonic scenario satisfies all observations to produce the preserved configuration of troughs and impact basins. It also places all events in a temporal order that agrees with the observed stratigraphic relationships, whereby the Divalia Fossae formed well after the Rheasilvia basin (*9*, *10*) and implying that the Saturnalia Fossae are younger than the Veneneia basin. This warrants a reassessment of the currently established global stratigraphy of Vesta, which is based on the assumption that the Divalia Fossae and Rheasilvia basin and by implication the Saturnalia Fossae and Veneneia basin share the same age, respectively (*2*, *6*, *34*–*36*). Such geologic evolution not only provides invaluable insights into understanding Vesta as our largest observable differentiated rocky asteroid but also serves as a case study for such processes that may have been common among other asteroids, protoplanets, and moons.

MATERIALS AND METHODS

Stress modeling of reorientation and despinning

To derive optimal solutions for TPW, despinning, and up-spinning, we use the observed gravity field (*25*). Given that rotational contributions to the gravity field predominantly occur at spherical harmonic degree-2, we use the long-wavelength gravity field while excluding degree-2 contributions to refine our best-fit Veneneia and Rheasilvia gravity models.

In our modeling approach, basin compensation is addressed by applying Love number theory (*37*), assuming the interior structure specified in table 3 in (*27*). As Vesta cools, an elastic lithosphere is expected to develop, likely playing a crucial role in supporting the topography of its large impact craters. The mean crustal thickness is ~24 km, based on the model of (*27*), and the lithospheric thickness could exceed this value. However, the thickness of Vesta's elastic lithosphere and its evolution over geological time are poorly constrained and depend on modeling assumptions and the region considered [e.g., (*26*)]. For our calculations, we adopt an elastic lithosphere thickness in the range of 5 to 20 km. Variations within this range have negligible effects on the best-fit solutions, as the resulting elastic deformation remains minimal, in agreement with the findings of (*27*).

The misfit to be minimized is expressed as

$$\chi^2_{\text{basins}} \equiv \sum_{n=3}^{n_{\max}} \sum_{m=-n}^{n} \left(\frac{\Phi_{nm}^{\text{OBS}} - \Phi_{nm}^{\text{basins}}}{\delta\Phi_{nm}} \right)^2 \tag{1}$$

Here, Φ_{nm}^{OBS} represents the observed gravity coefficients, $\delta\Phi_{nm}$ corresponds to the uncertainties, and $\Phi_{nm}^{\text{basins}}$ denotes the gravity coefficients associated with the basins. Changing the truncation degree $n_{\max}$ results in negligible alterations for $3 \leq n_{\max} \leq 6$. We adopt the basin profiles from (*22*) for Rheasilvia and Veneneia and determine their best-fit maximum rim heights, which allows us to solve for their gravity coefficients separately. The geometric centers of the basins are located at 69.3°S, 54.1°E for Rheasilvia and 56.3°S, 43.2°E for Veneneia (*9*). Given Vesta's rapid rotation, rotational effects could influence even-degree harmonics for $n > 2$. To evaluate the impact of these higher-order rotational contributions, we tested solutions including only odd-degree harmonics and found negligible changes to the best-fit results.

Using the obtained best-fit basin solutions, we use the degree-2 gravity field to refine our solutions for TPW, despinning, and up-spinning. In this context, our goal is to minimize the misfit, expressed as

$$\chi^2_{n=2} \equiv \sum_{m=-2}^{2} \left(\frac{\Phi_{2m}^{\text{OBS}} - \Phi_{2m}^{\text{basins}} - \Phi_{2m}^{\text{RB}}}{\delta\Phi_{2m}} \right)^2 \tag{2}$$

Here, Φ_{2m}^{RB} represents the gravity coefficients associated with a remnant rotational bulge resulting from TPW, in conjunction with either despinning or up-spinning (*21*).

Exploring the entire parameter space of the initial rotation rate, as well as the initial rotation pole latitude and longitude, we present the misfit as a function of the initial rotation pole colatitude and longitude in supplementary fig. S2. The analysis reveals the best-fit initial rotation rate to be 16% larger than the present value (an initial rotation period of 4.49 hours), and the corresponding optimal initial rotation pole location is identified as 87°N 41°E. The inferred despinning could be due to the Veneneia and Rheasilvia impacts or to other subsequent impacts. Since our best-fit solutions rely on the present-day gravity field as a constraint, distinguishing between these scenarios is not feasible.

Last, with the determined best-fit initial rotation rate and rotation pole location, we proceeded to calculate the corresponding global stresses, following the methodology outlined in (*28*). Stress solutions were calculated using a shear modulus of 40 GPa and Poisson's ratio of 0.25. This approach assumes a spherically symmetric, unperturbed state and a thin elastic lithosphere. To validate the thin elastic lithosphere assumption, we also applied the model described in (*38*), which does not rely on this approximation. For thicknesses of the elastic lithosphere between 5 and 20 km, the resulting stress field showed negligible changes. Greater lithospheric thicknesses that may have been present if the impact basins are considered to be geologically late, would lead to bigger differences in the stress solution. Moreover, while the assumption of spherical symmetry is retained, we expect only minor deviations from this simplification, as degree-2 contributions primarily dominate TPW stresses. The predicted stresses are primarily driven by despinning, which generates a stress field that is symmetric about the rotation axis. However, the inclusion of TPW introduces asymmetry, disrupting this rotational symmetry in the stress field.

Rock-mechanical assessment of stress field

The stress modeling predicts the maximum and minimum horizontal principal stresses on the surface of Vesta (Fig. 3A) with the vertical stress, i.e., the overburden, acting on the surface to be 0 MPa. Among these three principal stresses, the maximum principal stress (σ_1) and the minimum principal stress (σ_3) can be used to resolve the normal and shear stresses (σ_n and τ) on particular orientations (alpha, α), which represents the angle from the direction of σ_1 to the physical fracture plane, including a joint and fault plane. For all stress calculations, we follow the geologic sign convention, where compression is positive, and the maximum principal stress is the maximum compressive (or least tensile) stress. The stress equations are (*39*)

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$$\sigma_n = \left(\frac{\sigma_1 + \sigma_3}{2}\right) - \left(\frac{\sigma_1 + \sigma_3}{2}\right)\cos 2\alpha \quad (3)$$

and

$$\tau = -\left(\frac{\sigma_1 + \sigma_3}{2}\right)\sin 2\alpha \quad (4)$$

The rock strength of the lithosphere can be given by the Griffith criterion [e.g., (*40*, *41*)] in the tensile regime and Coulomb criterion (*42*) in the compressive regime. The Coulomb criterion is better suited than Byerlee's law for the tectonics of Vesta, because it relates loading conditions to frictional and cohesive properties of the rock volume, whereas Byerlee's law assumes no cohesive strength, and thus rubble-pile conditions at the surface of the lithosphere (*43*). Rock failure occurs when the tectonic stress, represented by σ_n or τ, from Eqs. 3 and 4, meets and exceeds the rock strength, given in the form of

$$|\tau| \geq \sqrt{4T_0(\sigma_n + T_0)} \text{ for } \sigma_n < 0 \text{ (i. e., Griffith criterion)} \quad (5)$$

and

$$|\tau| \geq C_0 + \mu\sigma_n \text{ for } \sigma_n > 0 \text{ (i. e., Coulomb criterion)} \quad (6)$$

where C_0 is cohesion, T_0 is the tensile strength, and μ is the coefficient of friction. We use the T_0 of 0.2 MPa and C_0 of 0.4 MPa (*30*), which are the typical values for the highly fractured lithospheric basaltic rock mass such as those expected on Vesta (*15*). Otherwise, the rock is stable with no failure predicted.

When rock failure is predicted, we assess the fracture type by their stress mode. If any normal stress component is tensile (Eq. 5), it then predicts jointing. Optimal joint orientation (α_{opt}) is perpendicular to the direction of σ_3 (maximum tension), and the permissible orientations range from wherever solutions from Eq. 5 are fulfilled at α in Eq. 3. If both the maximum and minimum horizontal principal stresses are tensile and exceed T_0, jointing can occur at all orientations. When these horizontal principal tensile stresses are similar in magnitude, there is no optimal orientation for jointing.

If all stress components are compressive, the fault type can be assessed by applying Andersonian theory (*44*). The optimal fault orientation, α_{opt}, is given by

$$\alpha_{opt} = 45° - \frac{\phi_f}{2} \quad (7)$$

where ϕ_f is the angle of friction. We use the ϕ_f of ~40°, given by $\mu = \tan\phi_f$, which is the typical value for the lithospheric pressure of Vesta (*15*, *29*) (i.e., $\mu = 0.85$ when the pressure is below 110 MPa). The permissible orientations of faulting range from wherever Eq. 6 is fulfilled at α in Eq. 4. Since faulting requires a fully compressive stress state (*44*), and the tensile strength of basaltic rock masses is much lower than the compressive strength (*30*), if the rock mass experiences both tensile and compressive stress components, jointing occurs before any faulting can take place.

For each 15° by 15° bin, we checked if failure is predicted. If so, we distinguish the fracture types predicted between joints (Griffith criterion in Eq. 5) and faults (Coulomb criterion in Eq. 6). We then determine the permissible orientations as a range of the α value from Eqs. 3 and 4, as well as the optimal orientation predicted by α_{opt} in Eq. 7. We compare these results with the orientation of the Divalia Fossae to see if there is a match in fracture type and orientation. The analytical solutions are provided in data S1.

Supplementary Materials

This PDF file includes:
Figs. S1 and S2
Legend for data S1

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